

CARMA: Online Cluster-Adaptive Eviction for GPTCache

Cluster-Adaptive Reuse and Miss-pressure Admission

Talya Yermiyahu

Final Project Report — September 2026

Executive result. I added an opt-in, CPU-only eviction policy to GPTCache that learns bounded semantic topics and fine-grained reuse cells online, adapts their quotas with decayed demand and miss pressure, and rejects weak one-shot admissions. The factory default and public policy interface remain LRU-compatible for valid existing inputs; shared normalization, validation, and cleanup were hardened. SQLite and FAISS deletion use vector-first cleanup, targeted rollback, and fail-closed reconciliation. On ten paired 10,000-request seeds at capacity 100, CARMA improved category-shift valid-hit rate by **4.885 percentage points** over the per-seed stronger LRU/LFU baseline (95% paired bootstrap CI [4.677, 5.088], Holm-adjusted $p = 0.0234$). It also passed stationary non-regression and synthetic weighted-savings-proxy gates. Five fresh SQLite/FAISS processes per policy showed worst ratios across seeds of $1.032\times$ LRU for post-embedding cache-path p95 latency, $0.997\times$ LRU for throughput, and $1.043\times$ LRU for sampled peak RSS. The reported synthetic “token-saving” result is a deterministic weighted-savings proxy, not a measurement of real model tokens.

Formal Gate 7 v5 ran all **15/15** fresh processes and retained **45,000** real-text ONNX requests across five previously unused seeds. Its canonical terminal auditor reports a claimable **PASS** with zero errors and zero warnings: every seed satisfies all six frozen latency, throughput, and RSS bounds. The independent semantic reconstruction reports **PASS_OBSERVED**: all 25,400 hits are same-concept, with zero cross-concept, labeled-negative, stale-candidate, or response-mismatch hits. An isolated ordinary re-audit reproduced the canonical adjudication byte-for-byte. Earlier v1–v4 outcomes remain immutable and nonclaimable.

Two other preregistered gates **failed**. CARMA returned 30,000/30,000 valid scan-return hits, but LFU already returned 29,998/30,000, leaving only +0.0067 percentage points rather than the required +5.0. Separately, strict QQP v2 calibration selected no threshold, so the held-out split was never evaluated. These negative and invalid results are preserved rather than converted into a favorable aggregate claim.

Software baseline	GPTCache commit <code>c59fb3a6152a4458b2a070ca183b61c4b614095f</code>
Primary paper	SCALM, arXiv:2406.00025v1
Feature branch	<code>feature/gate7-end-to-end</code>
Primary policy config	topic .70, cell .97, half-life 500, quota strength 1.0

1 Introduction and related work

Semantic caches store responses under vector representations of queries and reuse an answer when a later query is sufficiently similar. GPTCache provides a modular open-source implementation: embeddings, vector search, similarity evaluation, scalar/vector storage, and eviction can be configured separately [1]. At the selected commit, the in-memory manager offers global LRU, LFU, FIFO, and random replacement over opaque row identifiers. Those policies know recency or frequency, but not whether several residents cover one redundant fine-grained reuse region or whether a miss belongs to a growing topic.

SCALM motivates semantic structure as a cache-management signal. It analyzes MOSS and LMSYS conversations, derives hierarchical semantic patterns, ranks them using hit/token-saving evidence, and reports large relative gains over GPTCache baselines [2]. Its pattern discovery is corpus-oriented. The question here is deliberately different:

Can a bounded policy discover useful semantic structure *during* the request stream, without offline ranking, a learned model, token cost as an input, or a paid LLM call?

Conventional cache research supplies complementary mechanisms. 2Q uses a second-chance history to resist one-hit pollution [3]; LRFU spans recency and frequency through weighted history [4]; ARC adapts between recency and frequency lists [5]; and TinyLFU uses approximate frequency for admission [6]. CARMA’s decayed demand overlaps that adaptive recency–frequency tradition, while its bounded repeat evidence resembles 2Q/TinyLFU admission. Its distinguishing mechanism is online semantic organization with topic-level retention controls, not a claim that these conventional ideas are new.

This question leads to three engineering requirements. First, answer correctness must remain controlled by GPTCache’s retrieval threshold; a policy cluster may guide retention but may never declare a response correct. Second, all state must remain bounded by cache capacity rather than trace length. Third, eviction must remove scalar and vector records together so a rejected FAISS vector cannot shadow a valid neighbor at `top_k=1`.

Contribution and claim boundary

The contribution is CARMA, a deterministic online policy plus backward-compatible GPTCache integration, failure-safe cleanup, restart reconstruction, tests, and a reproducible evaluation pipeline. The historical eight-gate audit records Gates 1, 3, 5, 6, and 8 as pass, Gates 2 and 4 as fail, and its original precomputed-vector Gate 7 as pending. Later real-text ONNX attempts do not rewrite that history: formal v1–v4 are immutable invalid attempts, while formal v5 is a separate frozen successor. V5 completed its full 5-by-3 matrix and is a claimable **PASS** under its own preregistered six-check rule. The historical audit remains mixed (five pass, two fail, one pending), but the current full-path Gate 7 supplement has now passed; these two statements refer to distinct frozen evidence generations.

2 Extension design

2.1 State and online clustering

Each admitted entry stores a topic, a fine-grained semantic cell, insertion and last-hit ticks, and decayed hit mass. Topics retain normalized centroids, resident IDs, demand, and miss mass. Cells retain centroids, residents, support, and last observation. The bounds are $\min(64, C)$ topics and $2C$ cells for capacity C .

On a miss, the normalized embedding joins the highest-cosine topic above `topic_threshold`, or creates a bounded topic. The same rule assigns a cell using `cell_threshold`. Centroids use a normalized exponential moving average. Ties always choose the lowest stable state ID. Low-support empty cells remain temporarily as bounded ghost history, allowing a second similar miss to challenge an incumbent even if the first candidate was rejected.

One logical tick occurs per cache hit or inserted miss. A statistic x last updated at t_0 decays lazily as

$$x(t) = x(t_0) 2^{-(t-t_0)/H},$$

where H is `demand_half_life`; $H = \infty$ disables decay. Lazy decay keeps the common hit path constant-time.

2.2 Quotas, value, and admission

For topic c , recent miss pressure and quota weight are

$$p_c = \frac{m_c + 1}{d_c + 2}, \quad w_c = \max\{\epsilon, d_c p_c\}^q,$$

where m_c is miss mass, d_c is demand, and q is `quota_strength`. Largest-remainder apportionment produces nonnegative integer quotas that sum exactly to capacity. A resident e in cell s has retention value

$$V_e = \frac{\text{support}(s)}{\max(1, |s|)} + 0.25 \text{ hitmass}(e).$$

The first term rewards demonstrated semantic coverage and discounts redundant residents in the same cell.

Below capacity, every valid candidate is admitted. At capacity, a candidate is rejected when its decayed cell support is below 1.5 or its topic quota is zero. Otherwise it competes against the weakest eligible resident, either inside its topic or in the most over-quota donor topic. Admission requires candidate value at least 1.05 times victim value. Victim ties use value, last hit, insertion age, and stable ID representation in that order.

2.3 Complexity and intended trade-off

Hits are $O(1)$ apart from periodic quota refresh. Miss assignment is $O((T + S)D)$ for bounded topics T , cells S , and dimension D ; victim selection is $O(C)$. CARMA intentionally pays more policy CPU than LRU/LFU in exchange for semantic coverage. It should help under phase change and pollution, but need not dominate when a simple baseline already retains the working set.

3 Correctness and GPTCache integration

Here, correctness means implementation and storage integrity; it does not imply that semantic reuse is safe, which is evaluated separately and fails to qualify in Section 7.3. CARMA is opt-in; the factory default and public interface remain LRU-compatible for valid existing inputs. Shared vector normalization, validation, and cleanup were hardened. The factory accepts `policy="CARMA"` and binds the storage callback internally, avoiding a subtle failure mode in which a prebuilt custom eviction instance is not connected to `SSDataManager._clear`. Insertions pass normalized embeddings and persisted metadata through a backward-compatible hook.

Invariant	Enforcement and test evidence
Count/state bounds	Quotas sum to C ; randomized sequences assert no overflow. Topics, cells, ghosts, and custom-ID ordinals stay bounded; full registries fail closed.
Scalar/vector agreement	Victims are deleted from FAISS before scalar commit; targeted rollback preserves earlier tombstones and fails closed when reconciliation cannot succeed.
Exactly-once callback	Batch victims are deduplicated and filtered against final residents before the out-of-lock callback.
Malformed metadata	Missing, non-finite, zero-length, or wrong-dimensional embeddings fail safely at full capacity.
Restart	Existing rows are reconstructed with a read-only scalar peek; demand/hit mass intentionally relearn from zero.
Determinism	Topic, cell, quota, donor, victim, and ghost-pruning ties each have a documented stable ordering.

Table 1: Core safety properties exercised by unit, property, failure-path, and SQLite/FAISS integration tests.

The failure-path suite includes callback exceptions, unhashable and representation-colliding IDs, re-admission within one batch, vector-delete failure after scalar soft deletion, and close/reopen reconstruction. The implementation becomes explicitly unhealthy after unrecoverable cleanup rather than continuing with split internal/external state.

3.1 Validation layers

The project uses four layers: (1) focused CARMA unit/property tests; (2) relevant upstream GPTCaché eviction and SQLite/FAISS tests; (3) deterministic CI traces with expected schemas/digests; and (4) real SQLite plus FAISS runs in fresh child processes. Python 3.12.13 is the primary pinned target; an explicit Python 3.8 compatibility slice protects GPTCaché’s older supported syntax. No optional backend may trigger a runtime package installation during verification.

3.2 Restart limitation

Resident semantic structure is reconstructed from stored embeddings, but adaptive demand and hit mass are not persisted. They restart at zero and relearn online. Persisting them correctly would require a versioned metadata schema and migration rules across GPTCaché backends; this is future work, not silently simulated state.

4 Experimental setup

The protocol was frozen before the inferential test. Validation and test seeds are disjoint, and test results never enter parameter selection. The 96-point grid crosses topic thresholds $\{.70, .80, .90\}$, cell thresholds $\{.95, .97\}$, half-lives $\{100, 500, 2000, \infty\}$, and quota strengths $\{0, .5, 1, 2\}$. Three validation seeds and three workloads select the highest workload-normalized valid-hit rate subject to zero synthetic false hits. The selected configuration is $(.70, .97, 500, 1.0)$.

Workload	Frozen construction	Purpose
Stationary Zipf	10,000 requests, exponent 1.2, 15% one-shot	Steady skew and non-regression.
Category shift	Five 2,000-request phases; 80% demand moves	Adaptation to new hot topics and recovery lag.
Pollution scan	3,000 warm, 4,000 unique, 3,000 return	Scan resistance and working-set recovery.
MOSS novel-long	Longest quartile, token-bucket stratified, 200 unique	Exact-key LRU expected-miss control and recorded-response/token accounting; not a CARMA overhead comparison.
QQP safety	Component-disjoint calibration/test split	Human-labeled semantic precision and false-hit control.

Table 2: Primary and safety workloads. Synthetic primary runs use capacity 100 and ten paired seeds; real system runs use five fresh processes per policy (15 policy processes total).

A hit is *valid* only if the returned answer ID is allowed by the request’s ground-truth concept. False hits save zero valid tokens. Primary metrics are valid-hit rate (VHR), hit precision, false-hit rate, opportunity recall, and a synthetic weighted-savings proxy. In the synthetic catalog, the per-concept weight is the deterministic formula $16 + (31t + 17c + 13q) \bmod 241$; it is not a tokenizer count or a SCALM token measurement. MOSS separately records actual corpus tokens. System metrics include request latency percentiles, throughput, CPU time, sampled RSS, and storage invariants.

Paired deltas use 10,000-resample seed bootstraps. Confirmatory VHR comparisons against LRU, LFU, and the per-seed stronger baseline share an exact Wilcoxon Holm family. Capacity sweeps use the same capacity-100 traces at capacities 20, 50, 100, and 200. Ablations disable clustering, decay, admission, or quota-aware eviction separately.

The compact synthetic catalog uses cosine 1.0 for repeated requests to the same labeled concept, 0.96 for a deliberate distinct near-twin, 0.88 for other distinct concepts in the same source cell, 0.72 for concepts in different source cells of one topic, and a measured 0.00–0.12 range across topics. The range arises because compact cell-local feature blocks are reused across topics; it remains below the 0.70 topic threshold. At the selected 0.97 cell threshold, no two distinct synthetic concepts share a learned CARMA cell. Thus the primary synthetic gain cannot establish a paraphrase-cell aggregation benefit; it mainly tests decay, admission, and retention under a controlled geometry rather than natural-language semantic generalization.

5 Synthetic results

Workload	LRU VHR	LFU VHR	CARMA VHR	CARMA vs. stronger
Stationary	43.457%	47.514%	54.636%	+7.122 pp
Category shift	63.376%	52.181%	68.261%	+4.885 pp
Pollution scan (whole)	58.400%	59.198%	59.200%	+0.002 pp
Pollution return phase	97.333%	99.993%	100.000%	+0.0067 pp

Table 3: Ten-seed means at capacity 100. “Stronger” is chosen per seed and metric before forming paired differences.

Figure 1 summarizes the primary whole-trace comparisons.

Valid-hit-rate delta vs stronger baseline

Paired seed mean and 95% bootstrap CI; primary capacity 100.
inferential full protocol.

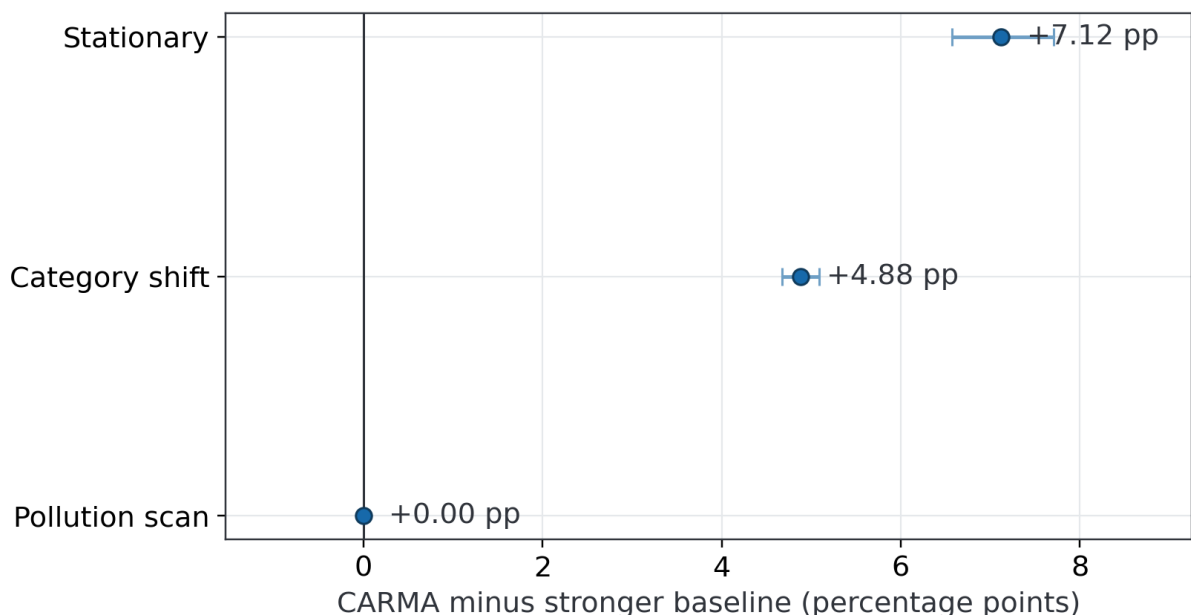


Figure 1: Primary whole-trace VHR difference between CARMA and the per-seed stronger LRU/LFU baseline. Error bars are paired 95% bootstrap intervals.

The category-shift result passes its preregistered gate: +4.885 pp, CI [4.677, 5.088], Holm-adjusted $p = 0.0234$. Stationary performance also passes the non-regression gate with CI lower bound +6.572 pp. The synthetic weighted-savings-proxy deltas pass in all workloads: stationary +9.503 pp [8.641, 10.443], shift +5.251 pp [5.009, 5.475], and pollution +0.0015 pp [0, 0.0037]. These values use the deterministic catalog weights above, not empirical token counts.

This is a substantive failed hypothesis, not a measurement failure. The supplementary plot at `artifacts/samples/analysis/scan-return.svg` draws the frozen +5.0 pp line. CARMA protects the original set, but LFU protects it almost equally well: the paired delta is +0.0067 pp, CI [0, 0.0167], Holm $p = 1.0$. In the separate exploratory 0.95-threshold hard-negative diagnostic, every policy produced one deliberate false hit per 200 requests (0.5% FHR), so the CARMA–baseline false-hit difference was zero. That diagnostic tests accounting and does not enter configuration selection or the primary threshold-0.97 claim.

6 Sensitivity and ablations

The labeled exploratory capacity curve is retained as `artifacts/samples/analysis/capacity-curve.svg`; its fixed-trace trend is summarized below rather than consuming another report page.

CARMA’s advantage is largest when capacity is constrained and under category shift. Averaging only for descriptive orientation across the three workloads, its VHR is 43.49%, 54.35%, and 63.97% at capacities 20, 50, and 200, respectively, compared with LRU’s 33.16%, 46.41%, and 60.74%.

Policy	Stationary	Category shift	Pollution scan
CARMA	54.636%	68.261%	59.200%
One global topic	54.983%	67.874%	59.200%
No decay	55.108%	59.980%	59.200%
No admission	53.025%	67.850%	55.388%
No quota eviction	54.983%	67.874%	59.200%

Table 4: Ablation mean VHR at capacity 100. These are descriptive component checks, not an additional confirmatory family.

Decay is the clearest adaptive component: removing it loses 8.281 points on category shift while slightly helping stationary traffic. Admission matters most on the scan workload; disabling it loses 3.812 points. In this synthetic geometry, replacing coarse topic clustering with one global topic changes stationary VHR by +0.347 pp and shift VHR by -0.387 pp relative to CARMA, while pollution is unchanged; fine-grained cell assignment remains enabled. The no-quota result is exactly identical to the one-global-topic result despite different topic counts. Coarse topic-clustering and quota effects are therefore not separately identifiable here; the results do not support a standalone claim that topic quotas preserve coverage. Decay and admission are the strongest observed drivers.

Recovery-lag evidence is weak: CARMA averages 1,808 requests versus the 2,000-request censoring limit for both baselines. Yet 31/40 CARMA and all 40 baseline transitions are censored; exploratory exact Wilcoxon $p = 0.0625$.

7 Real-system, MOSS, and semantic-safety evidence

The system benchmark uses the actual GPTCache `SSDataManager`, `SQLite`, and `FAISS` path, but replays precomputed vectors from the controlled synthetic catalog rather than running `paraphrase-albert-onnx`. Each policy runs in a fresh process on the same 3,000-request pollution trace, with immediate physical deletion and a final round-trip check of every active vector. Across five seeds, every run has zero false hits, zero stale candidates, and equal final scalar/vector counts.

Figure 2 separates policy-only work from post-embedding cache-path cost.

SQLite/FAISS latency and resource measurements

Precomputed vectors; embedding excluded. Descriptive per-seed means.
n=5 seed(s), mode=full; no across-seed Gate 7 rule is frozen.

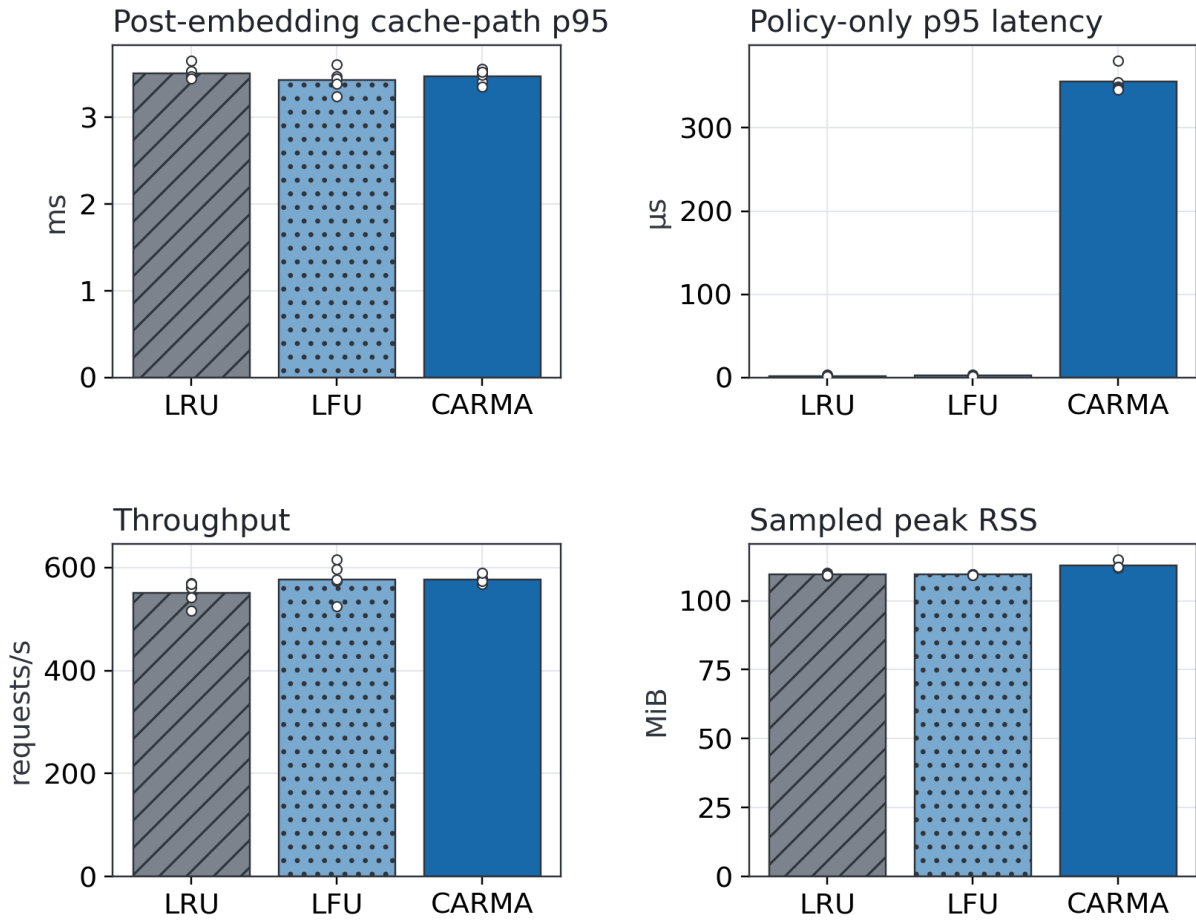


Figure 2: Real SQLite/FAISS measurements using precomputed synthetic vectors. Bars are five-seed means and circles are individual fresh-process runs. Policy-only overhead is shown separately from post-embedding cache-path request cost; embedding generation is excluded.

Policy	Mean	p50	p95	p99
LRU	1.804 ms	1.032 ms	3.511 ms	4.602 ms
LFU	1.724 ms	0.953 ms	3.433 ms	4.181 ms
CARMA	1.723 ms	0.875 ms	3.469 ms	3.952 ms

Table 5: Five-seed means of each fresh process’s request-latency summaries. The measured scope is the post-embedding cache path using precomputed vectors.

CARMA adds roughly 354 μ s to policy-only p95, reflecting bounded cosine assignment and victim selection. Post-embedding cache-path p95 nevertheless ranges from $0.936\times$ to $1.032\times$ LRU because policies create different hit, admission, and storage-cleanup mixes. The worst ratio across seeds is $0.997\times$ LRU for throughput and $1.043\times$ LRU (+4.8 MiB) for sampled peak RSS. Mean one-core CPU utilization is 96.9% for CARMA versus 94.1% for LRU. GPU use is not applicable to this CPU-only path, and macOS did not expose portable process I/O byte counters. Every seed satisfies the individual numerical checks; the historical Gate 7 entry remains pending because its contract omitted an across-seed aggregation rule and specified an ONNX integration embedding that this run did not use.

Only each run’s mean and quantiles were retained; individual request-latency samples were not retained in the integration artifacts. Figure 2 therefore shows the cross-seed distribution of p95 summaries, not a reconstructable per-request CDF. This is an artifact-granularity limitation despite collecting the frozen mean, p50, p95, and p99 metrics in process. The later v2 attempt does retain per-request samples, but its invalid ledger permits diagnostic use only.

7.1 Formal full-path ONNX lineage and v5 result

Gate 7 now has one historical diagnostic and five formal protocol versions (Table 6). The historical integration study used real SQLite/FAISS but precomputed synthetic vectors. Formal v1 stopped after one child because it conflated a semantic failure with structural corruption. Formal v2, v3, and v4 each completed a 5-by-3 matrix, but none is claimable. V5 is a separately frozen experiment that completed the same matrix and terminalized as a claimable **PASS**; it does not repair or reinterpret any earlier result.

Generation	Measured path	Status	Claim boundary
Historical	SQLite/FAISS with precomputed synthetic vectors	PENDING	Post-embedding diagnostics only; no frozen full-path ONNX adjudication.
Formal v1	Real QQP text, ONNX, one completed CARMA child	INVALID	Incomplete matrix and structurally conflated semantic failure.
Formal v2	Real QQP text, ONNX, all 15 fresh children	INVALID	Complete data, but seven preterminal errors and no terminal ledger event.
Formal v3	Real QQP text, ONNX, all 15 fresh children	INVALID	Complete data; own systems and semantic rules fail; contract mismatch leaves START unmatched.
Formal v4	Real QQP text, ONNX, all 15 fresh children	INVALID	Complete data; seven structural audit errors and descriptive numerical FAIL.
Formal v5	Fresh QQP traces, ONNX, all 15 fresh processes	PASS	Claimable complete matrix; all six checks pass on every seed; semantic guardrail is PASS_OBSERVED .

Table 6: Gate 7 lineage. Each generation is preserved under its own protocol; later evidence does not repair or reinterpret an earlier attempt.

The complete v2–v5 matrices measure the full `gptcache.adapter.adapter.adapt` path. It includes real text, a pinned 768-dimensional ALBERT ONNX embedding, similarity, SQLite, FAISS, policy and cleanup work, recorded miss response, and response return. LRU, LFU, and CARMA receive the same 3,000-request trace at capacity 100 in fresh child processes and storage, under a documented counterbalanced order. Each complete matrix has 45,000 request rows with mutually exclusive stage timings and external resource samples.

V2–v4 remain diagnostic only. V2 completed its matrix but had seven preterminal errors and no TERMINAL. V3 also completed but a manifest-contract shape defect blocked terminalization; under its own rules, both systems and semantic status fail. V4 introduced the six bounds retained by v5. Its first attempt was invalid because of

stale package metadata; its second completed all 15 processes but terminalized **INVALID** because of identity, warm-up, and resource-cadence control errors. Descriptively, every v4 seed exceeds the 0.5 ms paired policy-exclusive limit and one exceeds the post-embedding limit. These immutable records motivate v5 but receive no status change from it.

V5 was designed after observing v4 and therefore uses new seeds 20261201–20261205 and fresh trace hashes. It keeps the workload, timing boundaries, comparators, and all six numerical limits. The two identity checks now agree on path-bound producer records. Resource cadence requires at least 99.9% of gaps within 200 ms and imposes a separate 1 s hard ceiling. CARMA’s transaction snapshot now stores field tuples and reconstructs dataclasses only on exceptional rollback; quota refresh is deferred past a first-occurrence rejection that never reads quotas. A differential replay matched v4 admission and eviction decisions across 18,000 randomized events. The formal run was launched once from clean commit `f59d64d93a5599149538003078403f1b33e2bab1` and annotated tag `gate7f-onnx-v5-talya-formal-source`. It completed all 15 fresh policy-seed processes, 45,000 request rows, and 224,804 external resource rows. The canonical TERMINAL and ordinary auditor both report a claimable **PASS**, with zero errors and zero warnings. A second isolated invocation of the frozen ordinary auditor produced a byte-identical adjudication, SHA-256

`66378e671784b8f74a39910f918ef472c81ce36a1f1b9f4b10895430ff40b75a`.

All five seeds pass all six checks. Across seeds, the full-request p95 ratio is 0.999–1.090 against the 1.25 ceiling; post-embedding p95 delta is -0.205 – $+0.016$ ms against the $+0.500$ ms ceiling; paired policy-exclusive p95 delta is 0.225–0.239 ms against 0.500 ms; throughput ratio is 0.988–1.004 against the 0.90 floor; peak-RSS ratio is 1.001–1.006 against the 1.20 ceiling; and peak-RSS delta is 0.5–2.3 MiB against 64 MiB. Independent semantic reconstruction reports **PASS_OBSERVED**: 25,400/25,400 hits are same-concept, with zero cross-concept, labeled-negative, stale-candidate, source-response, or returned- response mismatches. These are observations on this frozen QQP workload, not a universal semantic-safety guarantee.

Gate 7 v5 per-seed checks against frozen bounds

Claimable **PASS** • 15 fresh real-ONNX processes • 45,000 requests • all 30 checks pass

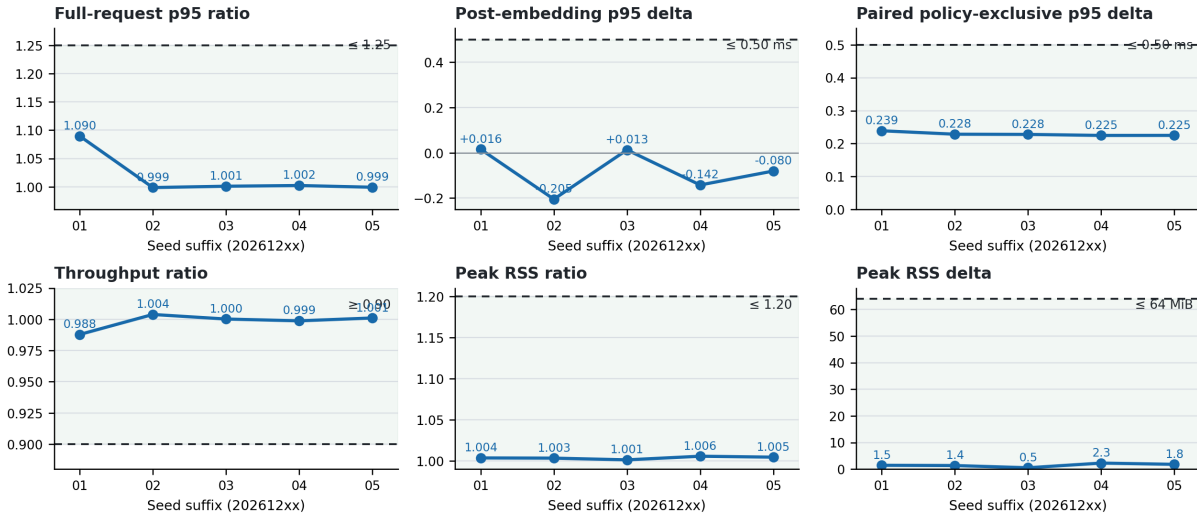


Figure 3: Formal Gate 7 v5 per-seed measurements against the six frozen bounds. Each seed contains 3,000 requests for each of LRU, LFU, and CARMA in fresh real-text ONNX processes (15 runs; 45,000 requests total). The dashed line in each panel is the applicable ceiling or floor; all 30 seed-level checks pass.

7.2 Pinned MOSS exact-key recorded-response replay

The CC-BY-4.0 MOSS archive [7] is fixed by repository revision and SHA-256. A full streaming CRC-verified pass covers 301,332 source conversations and 823,634 actual turns, then retains the lowest stable-hash 2,048-turn sample. The 200-request longest-quartile trace contains 557,243 context, prompt, and answer tokens. The replay uses an exact-key LRU cache. All requests are unique by design, so all 200 are misses, every recorded MOSS answer is delivered, and live model calls, cache hits, and false hits are all zero. Two runs produce byte-identical manifests, pool, requests, and CSV. This validates source and token accounting, but the lecturer’s novel-long

overhead criterion is only partial: this run neither compares CARMA/LRU/LFU semantic policies nor measures valid end-to-end long-prompt CARMA overhead.

The real archive also exposed 30,000 rows with `conversation_id=-1` and 2,727 declared-turn-count mismatches. Before producing a result, the parser was amended to hash canonical rows for negative sentinel IDs and to treat contiguous chat keys as authoritative while retaining both declared and actual counts. The final manifest records these anomalies and preserves the separate prepare manifest.

7.3 Held-out QQP safety

The component-disjoint calibration set contained 7,729 pairs: 5,652 positives and 2,077 negatives. None of the 20 frozen thresholds from 0.80 through 0.99 met the strict v2 selection rule. At threshold 0.96, precision is 0.97574893 and its one-sided 95% Wilson lower bound is 0.96420782; the false-hit rate is 0.00219951 and its one-sided Wilson upper bound is 0.00326719. Its Wilson precision lower bound is below the calibration-selection requirement of 0.99, and no other grid point qualifies, so the selected threshold is null. The separate downstream Gate 2 targets—held-out point precision at least 0.99, Wilson lower bound at least 0.98, and the CARMA false-hit-rate delta bound—were therefore never reached or evaluated. The protocol did not resolve held-out endpoints: all 56,963 held-out rows remain unevaluated, and no held-out precision or false-hit claim is made. Gate 2 is therefore **FAIL** regardless of the separate Gate 7 trace observations.

Gate	Status	Historical eight-gate adjudication
1	PASS	Exact-source host verifier passed; integration integrity failures are zero.
2	FAIL	No QQP calibration threshold met the precision prerequisite; held-out evaluation was not run.
3	PASS	Shift VHR +4.885 pp; CI [4.677, 5.088]; Holm $p = .0234$.
4	FAIL	Return VHR +0.0067 pp versus required +5.0 pp.
5	PASS	Stationary VHR delta CI lower bound +6.572 pp.
6	PASS	Synthetic weighted-savings-proxy delta CI lower bound meets every workload limit.
7	PENDING	The historical precomputed-vector study is diagnostic only; formal ONNX v1–v4 are separately INVALID . Formal v5 is separately PASS and claimable, but does not retroactively relabel this frozen row.
8	PASS	Two fresh network-isolated <code>linux/amd64</code> runs produced byte-identical benchmark artifacts and normalized logs.

Table 7: Historical success-gate table. Gates 2 and 4 remain failed; the original Gate 7 remains pending, while Table 6 records the later formal versions—including the claimable v5 **PASS**—without rewriting this audit.

8 Discussion, protocol audit, and limitations

8.1 What the experiments establish

CARMA’s strongest evidence is phase adaptation: decay prevents obsolete lifetime popularity from dominating after a topic shift, and admission protects scan performance. The ablations do not separately identify a benefit from clustering or quotas, so they cannot support a causal quota-coverage claim. The implementation cost is measurable in policy-only latency. All historical precomputed-vector seeds stay inside their diagnostic limits, but formal v1–v4 are operationally invalid; v4 also fails the six systems checks descriptively. Formal v5 supplies the missing claimable full-path result: it passes every frozen systems check on every seed and its observed semantic guardrail. The evidence therefore supports an adaptive-policy contribution driven most clearly by decay and admission, plus a bounded-performance result for the frozen v5 host, model, workload, and capacity.

The scan gate explains why raw perfection is not enough. CARMA achieves 100% return VHR, but the correct comparator is the stronger paired baseline, not LRU alone. LFU is already 99.993%; the prespecified five-point margin is impossible to claim from this trace. A future scan workload should make the hot set large enough, or interleave enough recurrent cold items, to distinguish coverage policies without changing the gate after observation.

8.2 Post-execution protocol conformance

The historical full bundle’s hashes, seed pairing, selected configuration, 480 aggregate rows, bootstraps, Wilcoxon tests, and Holm adjustment were independently recomputed exactly. That audit also found six deviations that this report does not hide:

1. the historical full bundle contains summary CSVs/metadata but not the promised per-request and process-resource JSONL, limiting reconstruction of individual phase events from that bundle alone;
2. synthetic policy execution order is fixed rather than randomized; state is isolated and timing is disabled, so this is unlikely to bias VHR;
3. synthetic false-hit intervals are seed bootstraps, not Wilson bounds; they must not be cited as the substantive semantic-safety interval; and
4. the preregistered scale sweep and secondary negative/ceiling controls were not run. Capacity sweeps and the four component ablations were run; and
5. the integration runs used precomputed 2,982-dimensional synthetic vectors rather than the frozen ONNX embedding, so their timing covers only the post-embedding cache path and Gate 7 remains diagnostic; and
6. the QQP implementation computed held-out cosine similarities transiently before the calibration abort, although it never selected a threshold, classified or aggregated held-out labels, exposed held-out metrics, or wrote those similarities.

The exact historical result remains tied to clean commit `28129f0b9785232827e741c0e2ff2bb96cc19423`; later tooling changes cannot retroactively alter its status. A separate deviation ledger documents impact and future-run remediation.

The final correctness and reproducibility evidence is tied to exact source commit:

`557c6ac0578cb6b77c5ae51595b49abdc0407e10`. The clean host and both containers used the same 527-file Git archive, SHA-256:

`51aea800e364c1d52bed566f48c4c49c1836d0d6a0b40b3bb8203867c14b58b4`. The report and curated evidence are the only changes in its single permitted direct packaging child. Gate 7 v1–v4 are separately bound to their clean sources, but inherit no pass from the host/container reproducibility gate. Their evidence and ledger outcomes remain frozen and may not be repaired in place. V5 therefore has a new contract, experiment ID, formal root, ledger schema, fresh seeds/traces, and source tag. Its sole eligible full attempt is bound to commit `f59d64d93a5599149538003078403f1b33e2bab1`; the hash-chained ledger contains GENESIS, START, and a PASS TERMINAL.

8.3 External validity

Synthetic vectors isolate eviction behavior but are not natural language. QQP calibration probes pairwise semantic matching; its component-disjoint held-out split was not evaluated, and neither represents conversational distribution shift. MOSS is an exact-key recorded-response negative control rather than a CARMA semantic-policy or long-prompt overhead comparison. Historical system timing excludes embedding; formal v1–v4 include real ONNX embedding but are invalid evidence, while formal v5 is claimable and includes the full request path. The v5 result still uses one Apple-arm64 machine, five seeds, capacity 100, one pinned CPU ONNX model, and SQLite/FAISS. Restart loses adaptive counters. These boundaries limit generalization but make each retained claim traceable.

9 Conclusion and future work

This project delivers a cleanly factored GPTCache feature branch, an online cluster-aware policy, robust cross-store deletion, focused and upstream tests, frozen paired experiments, real backend measurements, pinned natural-data checks, deterministic analysis, container/CI definitions, and an auditable report. A 5,180,400-evaluation final-source replay reproduced all four published synthetic CSV summaries byte-for-byte. The central hypothesis is only partly supported: CARMA is compelling under category shift and passes stationary VHR non-regression, but semantic safety fails at QQP calibration and its preregistered scan advantage over the strongest baseline fails. Full-path Gate 7 v1–v4 are invalid and nonclaimable. V4 completed all 15 children, but its retained measurements also fail the unchanged systems limits descriptively and its semantic guardrail fails. The separately frozen v5 matrix completed all 15 processes and is a claimable **PASS**: every seed passes all six systems bounds, with zero audit errors or warnings, and the observed semantic guardrail also passes.

The v5 implementation addresses the observed v4 control defects without altering old evidence: matching path-bound identities, exact warm-up schemas, a coverage-based resource-cadence rule with a hard ceiling, and ad-

versarial terminal/auditor tests. It also removes two unnecessary operations from CARMA’s first-occurrence rejection path while preserving decisions in an 18,000-event differential replay. Its estimands and all six numerical limits are unchanged from v4. The clean, source-tagged 5×3 real-ONNX execution used five fresh seeds and no selective reruns. Its canonical TERMINAL and independent ordinary re-audit agree exactly on **PASS**. Future work should replicate this result on additional hosts, capacities, embedding models, and conversational workloads rather than weakening the frozen gate. Public PR [zilliztech/GPTCache#696](https://github.com/zilliztech/GPTCache/pull/696) targets dev; it is mergeable and DCO-passing, but no maintainer has reviewed or adopted it. The exceptional adopted-PR route therefore remains an unmet external criterion.

Lecturer-criterion audit

Criterion	Status	Evidence
Correctness (40%)	Implementation PASS ; semantic FAIL	The v5 source host verifier passed 1,294 project tests with one optional skip, dependency integrity, and the deterministic CI benchmark. QQP calibration selected no qualifying threshold, so semantic reuse safety is not established for Gate 2; the distinct v5 trace guardrail is PASS_OBSERVED .
Reproducibility (30%)	PASS	Exact pins and one source archive produced matching host plus two fresh network-isolated container benchmark artifacts; evidence is checksum-linked; a final-source replay reproduced all four synthetic summaries byte-for-byte.
Performance (15%)	Mixed	Gates 3, 5, 6 pass; Gates 2 and 4 fail; each historical system seed meets its diagnostics; formal Gate 7 v1–v4 are invalid and v4 fails the unchanged systems rules descriptively; formal v5 is claimable PASS with all 30 frozen checks passing.
Clarity (15%)	Complete with caveat	Baseline rationale, policy/protocol docs, labeled figures, limitations, and this report are complete; the sealed README and contract expose the versioned v5 command and claim boundary.

Appendix: Artifact map

Code and tests are under `gptcache/manager/` and `tests/`; experiments and wrappers are under `benchmarks/` `carma/` and `scripts/`; course docs are under `docs/project/`; and curated checksummed evidence is under `artifacts/samples/`. Authoritative v1–v4 evidence remains immutable; compact checksum-bound preservation records for v2–v4 are tracked. The v4 root `artifacts/gate7-v4-onnx-attempts/` contains its two terminally invalid attempts. The v5 formal root `artifacts/gate7-v5-onnx-attempts/` contains the sole complete attempt, its hash-chained PASS ledger, manifest, run summaries, and canonical adjudication. Curated v5 values and Figure 3 are under `artifacts/samples/analysis/gate7-v5-pass/`. Environment/CI definitions are `Dockerfile.project`, `requirements-*.lock`, and `.github/workflows/carma-ci.yml`.

References

- [1] Bang Fu and Di Feng. “GPTCache: An Open-Source Semantic Cache for LLM Applications Enabling Faster Answers and Cost Savings.” *NLP-OSS 2023*, pp. 212–218. doi:10.18653/v1/2023.nlp-oss-1.24.
- [2] J. Li, C. Xu, F. Wang, I. M. von Riedemann, C. Zhang, and J. Liu. “SCALM: Towards Semantic Caching for Automated Chat Services with Large Language Models.” arXiv:2406.00025v1, 2024. arxiv.org/abs/2406.00025.
- [3] Theodore Johnson and Dennis Shasha. “2Q: A Low Overhead High Performance Buffer Management Replacement Algorithm.” *VLDB 1994*, pp. 439–450. vldb.org/conf/1994/P439.PDF.
- [4] Donghee Lee et al. “LRFU: A Spectrum of Policies that Subsumes the Least Recently Used and Least Frequently Used Policies.” *IEEE Transactions on Computers* 50(12), 2001, pp. 1352–1361. doi:10.1109/TC.2001.970573.
- [5] Nimrod Megiddo and Dharmendra S. Modha. “ARC: A Self-Tuning, Low Overhead Replacement Cache.” *FAST 2003. USENIX proceedings page*.
- [6] Gil Einziger, Roy Friedman, and Ben Manes. “TinyLFU: A Highly Efficient Cache Admission Policy.” *ACM Transactions on Storage* 13(4), Article 35, 2017. doi:10.1145/3149371.
- [7] OpenMOSS Team. “moss-003-sft-data,” revision 42e216d3e3fb331c18d5fa6e7cb4f1c53eef24a4, CC-BY-4.0. [dataset](https://openmoss.github.io/dataset).